Timing a possible encounter

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate orbital period from an encounter time.

SUCCESS CRITERIA - I can...

1. Use a clearly stated period and time reference.
2. Check positions of both bodies at the same time.
3. Explain why geometry and uncertainty matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “encounter” and “period” to explain your model or result. Sentence starters are optional.

encounter: objects reaching nearby positions at the same time

period: time for one complete cycle or orbit

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

At fixed mass and speed, circle radius becomes 3 times as large.
Compare acceleration and net force using equations.

Word help: net force = the vector sum of all forces on one object

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

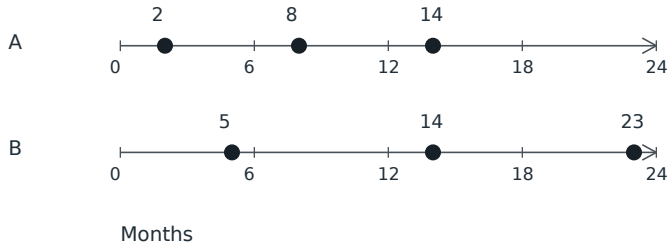
02 Worked example: explain the model

Reference: $t_{arrival} = t_{start}$ + travel time.
An encounter needs matching position and time. A period T is one complete orbit. Use the supplied fictional timing model.

Worked reasoning: Two objects pass the same crossing point: A at months 2, 8, 14 and B at months 5, 14, 23. Only month 14 is a shared time in these lists.

Your turn: Draw two timelines and connect the matching time.
Explain why a shared location alone is insufficient.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

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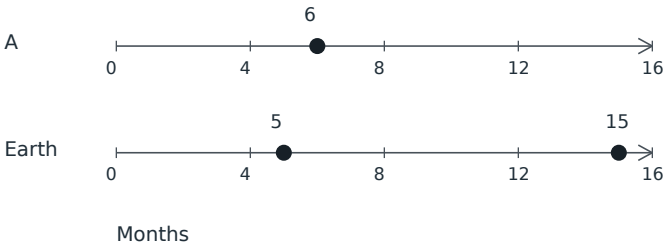
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A simplified orbit model has two paths that cross at the same point. Object A arrives in 6 months. Earth arrives in 5 months and again in 15 months.
Mark these times. Does sharing a crossing point establish an encounter?
Explain.

Sentence starter: Under the assumption …, the evidence … supports … because … .

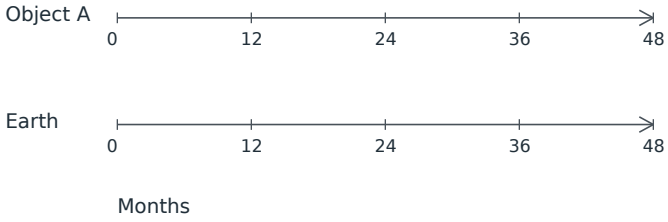


Marzano check - circle: 1 2 3

04 Apply the physics

Toy crossings: asteroid at months 6, 24, 42; Earth at months 6, 18, 30, 42.
Identify common crossing times and explain a limit of the model.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: 1 2 3

05 Explain your reasoning

A student turns a ±2-day timing range into “a 50% chance of impact.” Explain why that does not follow.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

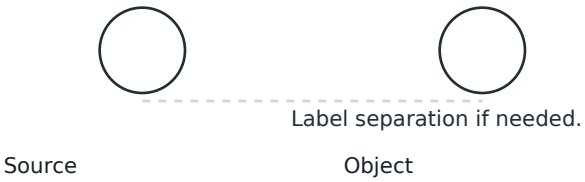
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

An object weighs 39.2 N where g=9.8 N/kg.
Find its mass and its weight in a field of 6 N/kg.
Label the masses, field or force relevant to the question.

Word help: weight = the gravitational force on an object

Sentence starter: I found … [unit]. This value means … in this situation.

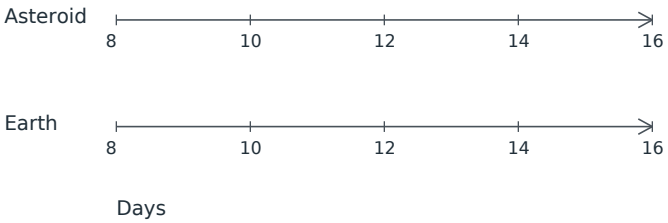


Marzano check - circle: 1 2 3

07 Exit check

An asteroid's crossing window is days 10-14; Earth's is days 13-15.
Find overlap and state what it does and does not establish.

Sentence starter: I found … [unit]. This value means … in this situation.

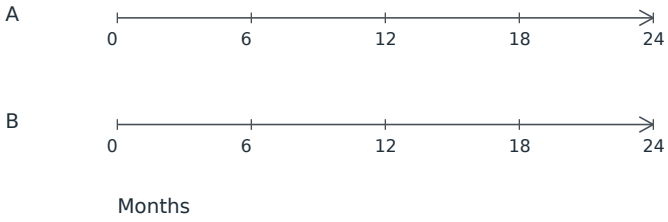


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A reaches a crossing at months 1+6 n; B at months 3+8 m, for nonnegative integers n, m.
Find their first common crossing time and a limit of this timing model.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3